Derivative Securities FNCE 30007

Tutorial 4

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Review

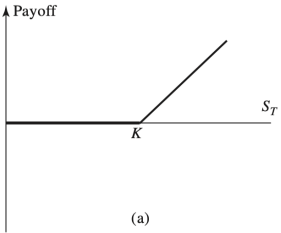
Basic types of option:

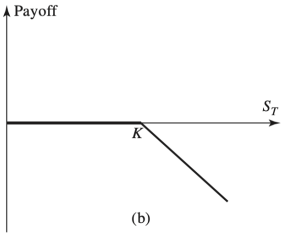
- Call option: the right for long position to buy or the obligation for the short position to sell an underlying asset during a particular period of time
- Put option: the right for long position to sell or the obligation for the short position to buy an underlying asset during a particular period of time

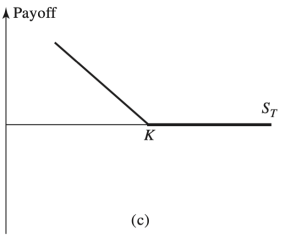
The buyer (holder) of the option pays the premium to the seller (writer) and gets the right to buy / sell the underlying asset from the seller during a particular period of time specified in the call / put option.

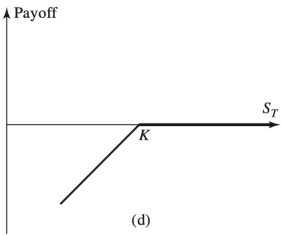
The seller (writer) sells an option contract to the buyer (holder), accepts a premium and assumes the obligation that is specified in the call / put contract

Review









Payoffs from positions in European options with strike price = K; price of asset at maturity = S_T.

(a) long call,

(b) short call,

(c) long put,

(d) short put.

Review

Moneyness, Intrinsic value, Time value

- **Intrinsic value:** max of zero & the value of option if exercised immediately
  - $\max(S - K, 0)$ for a call option
  - $\max(K - S, 0)$ for a put option
- **Moneyness:** for the call (put) option is
  - In the money if $S > K$ ($S < K$)
  - Out of the money if $S < K$ ($S > K$)
  - At the money if $S = K$($S = K$)

Option value = Intrinsic value + Time value
Time value:

- captures possibility option may increase in value due to volatility in the underlying asset
- positive, declines exponentially with time and reaches zero at maturity

Review

The European call option price upper bound (on non-dividend paying stock):

$$c \leq S_0$$

Otherwise, we can take the short side of the call option, and buy the stock right now. Then our date 0 profit will be $c - S_0$.

- If $S_T > K$, then we sell the stock for S_T and pay $S_T - K$ to the holder (Cash settled).
- If $S_T \leq K$, then the holder won't even exercise the option. If we sell the stock, then we will earn S_T at the contract expiration date.

Review

The European call option price lower bound (on non-dividend paying stock):

Portfolio		**Time** 0	**Time** T	
			$S_T \geq K$	$S_T < K$
A	1 European Call	c	$S_T - K$	0
	Cash	Ke^{-rT}	K	K
B	1 Share	S_0	S_T	S_T
Portfolio Value			A = B	A > B

Table: Call Option Lower Bounds

From the table, we note that the portfolio A (long one call + cash invested at risk-free rate) always ends up with no worse case than B (long one share), therefore, portfolio A should be sold more expensive than B at time 0:

$$A \geq B \implies c + Ke^{-rT} \geq S_0 \implies c \geq S_0 - Ke^{-rT}$$

$$\text{and also, } c \implies \geq 0 \quad c \geq \max(S_0 - Ke^{-rT}, 0)$$

Review

Put-Call Parity (on non-dividend paying stock):

Portfolio		**Time** 0	**Time** T	
			$S_T \geq K$	$S_T < K$
E	1 European Call	c	$S_T - K$	0
	Cash	Ke^{-rT}	K	K
F	1 European Put	p	0	$K - S_T$
	1 Share	S_0	S_T	S_T
Portfolio Value			E = F	E = F

Table: Put Call Parity

Therefore, portfolio E and F should have the same price: $c + Ke^{-rT} = p + S_0$.

Tutorial Question

Q1: What is a lower bound for the price of a six-month call option on a non-dividend-paying stock when the stock price is \$80, the strike price is \$75, and the continuously compounded risk-free interest rate is 10% per annum?

We know that a portfolio of one call and the PV of the strike is at least as good as the stock. It follows that the value of a call plus the PV of the strike is at least as valuable as the stock and therefore:

$$c \geq S_0 - Ke^{-rT}$$
$$c \geq 80 - 75 \times e^{-0.1 \times 0.5}$$
$$c \geq 8.66$$

A lower bound for the call is \$8.66.

Tutorial Question

Q2: What is a lower bound for the price of a two-month European put option on a non-dividend-paying stock when the stock price is \$58, the strike price is \$65, and the risk-free interest rate is 5% per annum?

Analogous arguments to Question 1, a portfolio consisting of one put plus one share has a payoff at least as good as the depositing the PV of the strike today. Therefore:

$$
\begin{aligned}
p &\geq Ke^{-rT} - S_0 \\
p &\geq 65e^{-0.05/6} - 58 \\
p &\geq 6.46
\end{aligned}
$$

A lower bound for the put is \$6.46.

Tutorial Question

Q3: Calls were traded on exchanges before puts. During this period of time when calls were traded but puts were not, how would you create a European put option on a non-dividend-paying stock synthetically? Assume no dividends are paid during the life of the contracts.

Put-call parity can be used to create a put from a portfolio call and a position in stock and cash markets. Recall our reviewing part.

We can

- Long a call, CF: $-c$.
- Invest cash into bank at risk-free rate, CF: $-Ke^{-rT}$
- Short sell one stock, CF: $+S_0$

This is equivalent to long a put, CF: $-p$.

Tutorial Question

Q4: The price of a European call that expires in six months and has a strike price of \$30 is \$2. The underlying stock price is \$29, and a dividend of \$0.50 is expected in two months and again in five months. Risk-free interest rates for all maturities are 10%. What is the price of a European put option that expires in six months and has a strike price of \$30?
We know that the following two portfolios have the same payoffs at expiry:

1. A European call plus cash equal to $D + Ke^{-rT}$ that earns the risk-free rate.
2. A European put plus one share, where D is the present value of the dividend.

Therefore the options must satisfy the put-call parity condition:

$$c + D + Ke^{-rT} = p + S_0.$$

Solving for p:

$$p = c + D + Ke^{-rT} - S_0$$
$$p = 2 + 0.5 \times \left(e^{-0.1/6} + e^{-0.1\times 5/12}\right) + 30e^{-0.1/2} - 29 = 2.51$$

Tutorial Question

Q5: Explain carefully the arbitrage opportunities in Question 4 if the European put price is \$3.

Action	**t=0**	**t=T=6 months**	
		$\mathbf{S_T > 30}$	$\mathbf{S_T \leq 30}$
Long call	-2	$S_T - 30$	0
Short put	3	0	$-(30 - S_T)$
Short stock	29	$-S_T$	$-S_T$
Invest Ke^{-rT}	$-Ke^{-rT} = -28.54$	30	30
Invest $PV(Div)$**, pay accordingly**	$-0.5e^{-10\% \times \frac{2}{12}} - 0.5e^{-10\% \times \frac{5}{12}} = -0.97$		
Outcome	0.49	0	0

Table: Portfolio Actions and Payoff at Time 0 and Time T

Note that at $T = 2/12$, $T = 5/12$, you take the invested cash to the one who lent you the stock. (Make sure he will not be worse off from lending you the stock).

Tutorial Question

Q6: Give an intuitive explanation of why the early exercise of an American put often becomes more attractive as the risk-free rate increases and volatility decreases.

- For an early exercise, it has to be the case that the put option is in-the-money.
- Suppose that the risk-free rate is quite high, then we are willing to cash out and invest at the risk-free rate.
- Furthermore, we are more willing to do so if the stock price's volatility is very low (since we are quite sure that even if the stock price decreases, we cannot be better-off too much from later exercise).

Tutorial Question

A summary of early exercise for American call and put options.

Call

- No Dividend: Never early exercise. Since one can always buy the stock at or lower than the strike price at the expiration date.
- With Dividend: if the dividend amount is significant and exercising the option early to capture the dividend is economically advantageous, the option holder may choose to exercise the option before its expiration date.

Put

- American put options may be exercised early when the underlying asset's price drops significantly and it becomes financially advantageous for the option holder to exercise the put option before its expiration date.
- Other cases: Q6. (risk-free rate increases/ volatility decrease)

Tutorial Question

Q7: After reading the GameStop case, think about the following questions before the tutorial.

7.1 How Would You Trade GameStop?

POV: It is late 2020. You believe that GameStop can successfully improve its business by expanding its online operations and changing its business model. If the turnaround is successful, you expect investors to become more optimistic and GME's share price to rise substantially. However, given the uncertainty surrounding the company, you do not want to risk losing the full amount required to purchase GME shares.

Which of the following option positions (Long Call, Long Put, Short Put, Short Call) would best reflect your market view and risk preference? Explain your choice.

Tutorial Question

Q7.1 Answer: Long Call

A long call is appropriate because the investor is bullish on GME and expects the share price to rise substantially.

- It provides upside exposure if the share price increases.
- The maximum possible loss is limited to the option premium paid.
- It allows the investor to benefit from a large upside move without committing the full amount required to purchase the shares.

This position therefore matches both the bullish market view and the preference for limited downside risk.

Tutorial Question

7.2 American Put–Call Parity in Current Market Conditions

For this question, use the current (August 2026) market data provided in the second worksheet (Tutorial - GME 2026) of *GameStop_Options_Case_FNCE30007.xlsx*.

(a) Using the data provided, calculate the lower and upper bounds for the American put option implied by the American put–call parity relationship.

(b) Compare the observed market put price with your calculated range. Does the observed put price fall within the no-arbitrage range?

(c) Compare your result with the January 2021 GameStop short-squeeze example discussed in the lecture. What do you notice, and what might explain the difference?

Tutorial Question

Q7.2 Answer: American Put No-Arbitrage Range

For a non-dividend-paying stock, the American put price should satisfy:

$$c + Ke^{-rT} - S_0 < p < c + K - S_0.$$

Using $S_0 = 18.79$, $K = 19.00$, $c = 1.86$, $r = 3.84\%$, and $T = 0.1781$:

$$\text{Lower bound} = c + Ke^{-rT} - S_0 = 1.94,$$
$$\text{Upper bound} = c + K - S_0 = 2.07.$$

Therefore, the American put no-arbitrage range is:

$$\boxed{\$1.94 < p < \$2.07}.$$

Tutorial Question

Q7.2 Answer: Market Comparison

The observed market put price is $p = \$2.00$. Since

$$\$1.94 < \$2.00 < \$2.07,$$

the observed put price falls within the American put no-arbitrage range for the 2026 data.

During the January 2021 short squeeze, GME experienced extreme volatility, severe short-selling constraints, and substantial trading frictions. These conditions made arbitrage more difficult and costly, allowing larger pricing inconsistencies to persist.

By contrast, the 2026 market conditions are relatively normal. Arbitrage and short-selling mechanisms can operate more effectively, helping keep stock, call, and put prices more closely aligned with no-arbitrage relationships.